Alright — this is where we lock it in so Chapter 5 basically writes itself.

What I'm giving you now is **not prose**, it's a **production blueprint** you can follow step-by-step. If you stick to this, you won't drift, you won't forget evidence, and you won't undercook any section.

🧱 Chapter 5 — Drafting Blueprint (Final)

🎯 Target length

- **Total**: ~2,500–3,000 words
- Bias toward **figures + tables over text**

🧠 5.1 Evaluation Framework and Scope

🎯 Target: 200–300 words

Purpose

Define:

- what you evaluate
- what you don't
- how evidence is structured

Content blocks

1. Opening (2–3 sentences)

- This chapter evaluates IT4s as:
  - interactive system
  - closed-loop architecture
  - planner-driven system

---

2. Evaluation framing

Say explicitly:

- not evaluating:
  - perceptual musical quality
  - user study / co-creativity
- evaluating:

    - behaviour
    - structure
    - responsiveness
    - real-time execution

---

3. Evaluation questions

📊 Table 5.1 — Evaluation Questions and Evidence

Claim	Question	Evidence	Section
Reactivity	Does output change with input?	Variant analysis	5.3
Interpretability	Are decisions explainable?	Planner margins	5.3
Structural validity	Are outputs distinct?	Skeleton metrics	5.4
Real-time	Is latency acceptable?	Timing logs	5.5
Fidelity	Does output match plan?	Plan vs Output	5.6

---

4. Evidence types (bullet list)

- unit tests
- batch datasets
- debug tools
- timing logs

⚠️ Rules

- No results
- No interpretation
- Just framing

🧠 5.2 Subsystem Correctness

🎯 Target: 250–350 words

Purpose

Establish:

> the system works → evaluation is valid

Content blocks

1. Pipeline recap (VERY short)

Just 1–2 sentences:

- capture → compile → analyse → plan → generate → playback

---

2. Subsystem verification

📊 Table 5.2 — Subsystem Evaluation Summary

Subsystem	**Evidence**	**Verified Behaviour**
Capture	tests	correct segmentation
Compiler	inspection	correct quantisation
Analyser	tests	correct features
Planner	tests	deterministic + interpretable
Generator	tests	structural constraints enforced
Loop	debug	closed-loop execution

---

3. Short conclusion

> System is stable enough for behavioural evaluation

⚠️ Rules

- No deep analysis
- No repeating Chapter 3
- Keep it tight

🧠 5.3 Planner Behaviour, Interpretability, and Reactivity

🎯 Target: 600–700 words (CORE SECTION)

🔥 Structure

5.3.1 Response Distribution

📊 Figure 5.1 — Planner Distribution

(bar chart)

Use F1:

- Intensify: 39.33%
- Fill/Mirror: 20.67%
- Complement/Contrast lower

Write:

- non-degenerate behaviour
- structured bias (NOT “balanced”)
- intensify dominance

5.3.2 Decision Confidence

📊 Figure 5.2 — Margin Distribution

Use F1 margins:

- 0 → ambiguous
- high → confident

Key point

> planner exposes uncertainty

5.3.3 Reactivity (Variant Analysis)

📊 Figure 5.3 — Variant Change Rate

Use F3:

- 23.33% changed
- feature-dependent variation

Key interpretation

- not rigid
- not unstable

5.3.4 Transition Behaviour

Use:

- Fill → Intensify
- Mirror → Intensify
- Complement → Mirror

Key insight

> structured decision boundaries (not random)

🧠 Section conclusion

> planner is interpretable, structured, and input-sensitive

🧠 5.4 Structural Response Distinctiveness

🎯 Target: 600–700 words (CORE SECTION)

🔥 Structure

5.4.1 Aggregate Structural Profiles

📊 Table 5.4 — Skeleton Metrics

Use F2:

- overlap vs gap-fill patterns
- anchor behaviour

Write:

- Mirror/Simplify → source-aligned
- Complement/Contrast → gap-oriented
- Intensify/Fill → expansion

5.4.2 Pairwise Divergence

📊 Figure 5.5 — Divergence Heatmap

Use:

- Simplify most distinct
- Fill/Intensify closest

🔥 Key insight

> structural clustering emerges naturally

5.4.3 Case-Level Examples

📊 Figure 5.4 — Skeleton Comparisons

Use:

- 2–3 curated cases

🔥 MUST INCLUDE INSIGHTS

Insight 1

> identical metrics ≠ identical structure

Insight 2

> Simplify = structural compression

Insight 3

> behaviour clusters emerge

🧠 Section conclusion

> response types are structurally meaningful but context-dependent

---

🧠 5.5 Closed-Loop Execution, Timing, and Observability

🎯 Target: 350–450 words

🔥 Structure

5.5.1 Closed-Loop Behaviour

📊 Figure 5.6 — Debug View

Show:

- phases
- loop state

Write:

- full loop exists
- system is observable

5.5.2 Timing Analysis

📊 Figure 5.9 — Latency Breakdown

Use F5:

- mean ≈ 2.6 ms
- max ≈ 66 ms

🔥 CRITICAL LINE

> use **post-capture latency**, not total latency

5.5.3 Stage Breakdown

Highlight:

- generation dominates
- others negligible

🧠 Section conclusion

> system is real-time, generation is bottleneck

🧠 5.6 Response Realisation Gap and Technical Limitations

🎯 Target: 500–600 words (HIGHEST MARK SECTION)

🔥 Structure

5.6.1 Plan vs Output Fidelity

📊 Figure 5.8 — Plan vs Output

Use F4:

- density error: 0.048
- gap mismatch: 0.149
- anchor mismatch: 0.578

🔥 Interpretation

- density ✓
- gap ⚠️
- anchors ❌

5.6.2 Response-Type Breakdown

- Mirror → strong alignment
- Complement → good
- Intensify / Fill → severe mismatch

🔥 CORE CLAIM

> planner intent ≠ realised output

5.6.3 Root Cause

Connect:

- FeatureTransformer
- generator not enforcing constraints

5.6.4 Broader Limitations

- no perceptual validation
- no tempo adaptation
- no expressive timing
- no user study

📊 Table 5.5 — Claim Strength

Claim	Status
Planner works	Strong
Structure valid	Strong
Real-time	Strong
Output fidelity	Partial
Interaction quality	Partial

🧠 Section conclusion

> system stronger as architecture than final performer

🧠 5.7 Evaluation Synthesis

🎯 Target: 200–300 words

Structure

1. Strong claims

- planner ✓
- structure ✓
- real-time ✓

---

2. Partial claims

- output fidelity ⚠️
- musical plausibility ⚠️

---

3. Deferred

- user studies
- ML realiser
- co-creative evaluation

Final sentence (VERY important)

> IT4s successfully demonstrates an interpretable, closed-loop, real-time interaction architecture, while highlighting the challenge of translating structured planning into fully realised musical response.

🚀 Final Checklist (DO THIS)

Before writing:

✓ You have:

- all CSV outputs
- all figures ready
- curated case studies
- debug screenshots

✓ You MUST:

- lead with figures
- reference findings (F1–F5)

- avoid repetition
- avoid vague claims

🧠 Final advice (real talk)

If you follow this:

- Chapter 5 will feel **tight, technical, and confident**
- not essay-y
- not waffle
- not defensive

And most importantly:

> It will read like you **understand your system deeply**, not like you're trying to impress someone.

If you want next:

👉 we can start writing **5.1 + 5.2 together (clean, fast, locked tone)** 👉 or jump straight into **5.3 (the heavy hitter)**